

Solutions to Math 124 Winter 2024 Final

1. (a) $\lim_{t \rightarrow 2} \frac{3^t - 9}{t - 2} =_{\text{LH}} \lim_{t \rightarrow 2} \frac{\ln 3 \cdot 3^t}{1} = 9 \ln 3$

(b) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + x + 1}}{\sqrt{7x^2 + 3}} = \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{1}{x} + \frac{1}{x^2}}}{\sqrt{7 + \frac{3}{x^2}}} = \frac{1}{\sqrt{7}}$

(c) $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\tan(3x)} =_{\text{LH}} \lim_{x \rightarrow 0} \frac{5 \cos(5x)}{3 \sec^2(3x)} = \frac{5}{3}$

2. (a) $f'(x) = \arctan(x^2 + 1) + x \cdot \frac{2x}{1 + (x^2 + 1)^2}$

(b) $g'(x) = \frac{e^{e^x} \cdot e^x \cdot \ln x - \frac{e^{e^x}}{x}}{(\ln x)^2}$

(c)

$$y = x^{\sqrt{x}+1}$$

$$\ln y = \ln(x^{\sqrt{x}+1}) = (\sqrt{x} + 1) \ln x$$

$$\frac{y'}{y} = \frac{1}{2\sqrt{x}} \ln x + (\sqrt{x} + 1) \cdot \frac{1}{x}$$

$$y' = \left(\frac{\ln x}{2\sqrt{x}} + \frac{\sqrt{x} + 1}{x} \right) x^{\sqrt{x}+1}$$

3. (a) $f'(2) = 1$

(f) $(-2, 0) \cup (1, 4)$

(b) $f''(0) = -1$

(g) $x = 0, 1, 3$

(c) $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = f'(0) = 0$

(h) $(1, 2) \cup (3, 4)$

(d) $\lim_{h \rightarrow 0} \frac{f(2+h) - f(2)}{h} = f'(2) = 1$

(i) $x = 1, 2, 3$

(e) $\lim_{x \rightarrow 3^+} \frac{f'(x)}{x - 3} = \lim_{x \rightarrow 3^+} \frac{f'(x) - f'(3)}{x - 3} = 2$

4. Implicit Differentiation:

$$3(p - 20)^2 \frac{dp}{dq} + \frac{dp}{dq} q + p + 3q^2 = 0.$$

When $q = 16$, $p = 5$

$$3(-15)^2 \frac{dp}{dq} + \frac{dp}{dq} \cdot 16 + p + 3 \cdot 16^2 = 0.$$

so $\frac{dp}{dq} = -\frac{773}{691}$ and the tangent line is

$$p - 5 = -\frac{773}{691}(q - 16)$$

so the approximation is

$$p - 5 \approx -\frac{773}{691}(17 - 16)$$

so $p \approx 5 - \frac{773}{691} \approx 3.88$

5. (a) Solving

$$0 = y = (t - 2)^3 - 3(t - 2) = (t - 2)(t^2 - 4t + 1)$$

we have $t = 2$ or $t = 2 \pm \sqrt{3}$.

(b)

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3(t - 2)^2 - 3}{2(t - 2)}$$

When $t = 2 - \sqrt{3}$,

$$\frac{dy}{dx} = \frac{3(-\sqrt{3})^2 - 3}{2(-\sqrt{3})} = -\sqrt{3}$$

so the tangent line is at the point $x = (2 - \sqrt{3} - 2)^2 = 3$ and $y = (2 - \sqrt{3} - 2)^3 - 3(2 - \sqrt{3} - 2) = 0$ is

$$y = -\sqrt{3}(x - 3)$$

(c) $y = \sqrt{3}(x - 3)$.

6. Given $\frac{dV}{dt} = 5 - 3 = 2$, we want $\frac{dh}{dt}$. If r is the radius of the surface of the water, from similar triangles

$$\frac{2}{6} = \frac{r}{h}$$

or $r = h/3$. So the volume of water is

$$V = \frac{\pi r^2 h}{3} = \frac{\pi (h/3)^2 h}{3} = \frac{\pi h^3}{27}$$

and

$$\frac{dV}{dt} = \frac{\pi h^2}{9} \frac{dh}{dt}$$

so $\frac{dh}{dt} = \frac{2}{\pi}$ meters per minute.

7. Let x be the distance from the first pole to where the wire is connected to the ground. Then, the length of the wire is

$$f(x) = \sqrt{20^2 + x^2} + \sqrt{(60 - x)^2 + 40^2}$$

and

$$f'(x) = \frac{2x}{2\sqrt{20^2 + x^2}} + \frac{-2(60 - x)}{2\sqrt{(60 - x)^2 + 40^2}} = 0$$

when

$$x\sqrt{(60 - x)^2 + 40^2} = (60 - x)\sqrt{20^2 + x^2}$$

or

$$x^2((60 - x)^2 + 40^2) = (60 - x)^2(20^2 + x^2)$$

so $x = 20$ (or $x = -60$ which does not make sense). Checking endpoints and the critical point

$$f(0) \approx 92.1, \quad f(20) \approx 84.9, \quad f(60) \approx 103.2$$

so the minimum amount of wire is approximately 84.9 meters. (You can also verify the sign change of f' at $x = 20$ to show it must give the minimum length of wire.)

8. (a) When $y = 0$, the x -intercept is at $x = 1$. The y -intercept is at $y = f(0) = 1$.
 (b) The domain is all numbers so there is no vertical asymptote.

$$\lim_{x \rightarrow \infty} e^x(x-1)^2 = \infty$$

and

$$\lim_{x \rightarrow -\infty} e^x(x-1)^2 = 0$$

so $y = 0$ is the horizontal asymptote.

(c)

$$f'(x) = e^x(x-1)^2 + 2e^x(x-1) = e^x(x^2 - 1)$$

so the critical numbers are $x = \pm 1$. The function is increasing on $(-\infty, -1) \cup (1, \infty)$ and decreasing on $(-1, 1)$ so it has a local max at $x = -1$ and a local min at $x = 1$.

(d)

$$f''(x) = e^x(x^2 - 1) + e^x(2x) = e^x(x^2 + 2x - 1) = 0$$

when $x = -1 \pm \sqrt{2}$ which give points of inflection because f is concave up ($f'' < 0$) on $(-\infty, -1 - \sqrt{2})$ and $(-1 + \sqrt{2}, \infty)$ and concave down ($f'' > 0$) on $(-1 - \sqrt{2}, -1 + \sqrt{2})$.

(e) Graph of $y = f(x)$

